PROPOSAL DOCUMENT

Project: Java AFK NaNo
Target: Android (Kotlin JVM)
Date: November 2026
Status: Ready for Review

Java AFK NaNo v4.0 Evolution

Next-Generation Feature Concepts, UI/UX Redesign Blueprint & Engine Architecture

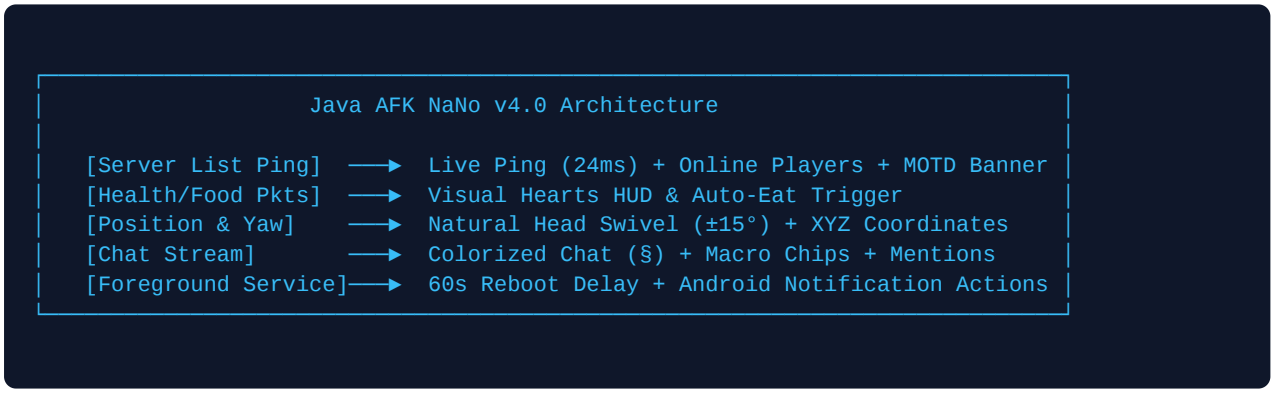
1. Executive Summary & Product Vision

Java AFK NaNo is an ultra-lightweight, battery-efficient headless Minecraft Java Edition client designed specifically for mobile Android devices. It connects directly to Minecraft servers (protocols 1.8 through 1.21.4+), authenticates, handles network protocol keep-alives, and maintains long-running AFK sessions using a foreground Android service.

The core philosophy of the application is the **"NaNo" standard**:

- **Sub-3 MB APK Size:** Pure Kotlin/JVM with zero WebView, Chromium, or JNI rendering baggage.
- **Near-Zero Battery Impact:** Consumes <0.5% battery over an 8-hour overnight AFK session.
- **Pure Headless Protocol:** Receives only network frames; never renders expensive 3D voxels.

The v4.0 upgrade aims to elevate Java AFK NaNo from a simple connection tool into an **intelligent, self-preserving Minecraft companion** with visual world cues, anti-detection behavior, and survival safeguards.



2. UX & UI Areas Requiring Redesign

1. Server Cards (Home Screen)

Current Issue: Cards are static. You cannot tell if a server is online or what its latency is without entering the session.

v4.0 Redesign:

- **Live Ping Pill:** Displays TCP latency (e.g., **24ms**) and player count (**18/50**).
- **1-Tap Card Toggle:** Instant Connect/Disconnect power button right on the card.
- **MOTD Strip:** Single-line server announcement preview.
- **Pulsing Alive Indicator:** Retains smooth 2s breathing glow ring for online bots.

2. Session Screen HUD

Current Issue: Top status card occupies 40% of the screen height, squeezing logs/chat into a tiny scrollable area.

v4.0 Redesign:

- **Minecraft Hearts HUD:** 10 authentic pixel-perfect red hearts & 10 food drumsticks.
- **Compact Metrics:** 3-column chip row for Coordinates, AFK Time, and Data Usage.
- **Expanded Chat Space:** 75% of screen allocated to streaming chat and logs.

3. Quick Macro Chips Bar

Current Issue: Executing frequent commands (**/afk**, **/spawn**, **/home**) requires opening the soft keyboard and typing manually every time.

v4.0 Redesign: Horizontally scrolling chips directly above the input box for 1-tap command execution.

4. Segmented Settings Sheet

Current Issue: Monolithic scroll dialog with raw text inputs.

v4.0 Redesign: Clean Material 3 Modal Bottom Sheet organized into 3 clear tabs: *Connection*, *AFK Automation*, and *Survival & Safety*.

3. Comprehensive 16-Feature Proposal Matrix

🛡️ Pillar 1: Pro AFK Behaviors & Survival Guard

1. Natural Camera Swivel (Yaw/Pitch Oscillation)

Anti-AFK | Effort: Low

Gently rotates the bot's camera angle by ±5° to ±18° with randomized intervals (every 12–28 seconds). Fools advanced anti-cheat plugins (Vulcan, GrimAC, Matrix) looking for static robotic angles. Network cost: 1 tiny packet (12 bytes) every 20 seconds; 0% battery impact.

2. Auto-Eat & Starvation Guard

Survival | Effort: Medium

Reads live food packets (`food = r.readVarInt()`). When hunger drops below 14 (7 drumsticks), the bot automatically sends an interaction packet on the hotbar food slot to eat, preventing starvation deaths while AFK at mob farms.

3. Sneak / Un-sneak Micro-Pulse

Social | Effort: Low

Transmits an `ENTITY_ACTION` packet (sneak for 400ms, then stand) periodically. Visually signals to other players on the server that the bot account is alive.

4. Emergency Disconnect on Damage

Safety | Effort: Low

Configurable threshold (e.g., < 4 hearts): If the bot takes damage from lava, suffocating blocks, or hostile mobs, it cleanly disconnects to preserve equipment and inventory.

🎮 Pillar 2: Visual HUD & Chat Superpowers

5. Authentic Minecraft Hearts & Food Drumsticks HUD

HUD | Effort: Low

Renders 10 red hearts and 10 food drumsticks in Jetpack Compose, updating dynamically whenever `PLAY_S2C_UPDATE_HEALTH` is received.

6. Quick Command Macro Chips Bar

Chat | Effort: Low

Horizontally scrolling chips (`/afk`, `/spawn`, `/home`, `/repair`, `/heal`, `/tpa yes`) above the chat input box for instant execution.

7. Minecraft Color-Coded Chat (§ Formatting)

Visual | Effort: Medium

Converts standard Minecraft section formatting codes (§a green, §c red, §6 gold, §l bold) into Compose `AnnotatedString` for authentic in-game chat reading.

8. Live XYZ Coordinates & Cardinal Compass

HUD | Effort: Low

Displays current coordinates (e.g. `124, 68, -312`) and cardinal orientation (North/South/East/West) parsed from player position packets.

🔔 Pillar 3: Background Reliability, Reconnects & Alerts

9. Intelligent Server Restart Grace Period — Reliability — Effort: Low

When a disconnect reason matches "Restarting" or "Rebooting", the client enters a 45–60s grace delay instead of immediate rapid reconnect backoff, preventing connection firewall bans.

10. Chat Mention & Direct Message Push Notifications — Alerts — Effort: Medium

Vibrates the phone and posts an Android heads-up notification when your username is mentioned or someone whispers via `/msg`.

11. Notification Action Buttons ([Stop All], [Reconnect]) — System — Effort: Low

Interactive action buttons attached directly to the ongoing Android foreground service notification.

12. Night AFK Scheduled Window — Automation — Effort: Medium

Allows scheduling connection windows (e.g. 11:30 PM to 6:30 AM) to save battery and mobile data.

🧰 Pillar 4: Multi-Bot Management & Presets

13. JSON & QR Code Server Config Backup/Share — Utility — Effort: Low

Export configured servers and command macros to clipboard JSON or share with a friend via a QR code.

14. 1-Tap Popular Server Presets — Presets — Effort: Low

Instant presets for Hypixel, 2b2t, DonutSMP, and Aternos with automated protocol detection.

15. Batch Master Actions (Start All / Stop All) — Batch — Effort: Low

Master controls in the drawer or top bar to connect or disconnect all configured bots in sequence.

16. Per-Server Custom Accent Color & Tags — Customization — Effort: Low

Tag servers (e.g. *SMP*, *Anarchy*, *Hub*) with custom badges for visual clarity.

4. Technical Implementation & Feasibility Matrix

Feature	Target Code Component	Network / System Mechanism	Overhead
Natural Head Swivel	`BotSession.kt`	`scheduleSwivel()` sending `sendLook()`	12 bytes / 20s (0% CPU)
Auto-Eat Food	`BotSession.kt`	Check `food < 14` in health packet → send item use	Event-driven (0 polling)
Minecraft Hearts HUD	`MainActivity.kt`	Jetpack Compose Row rendering heart icons	0 KB assets, pure UI
Quick Macro Bar	`MainActivity.kt`	Compose LazyRow calling `session.sendChat()`	Minimal Compose state
Live Ping & Player Count	`ServerPinger.kt`	Measure roundtrip TCP handshake + parse status JSON	One-shot async coroutine
Emergency Disconnect	`BotSession.kt`	Trigger `stop()` if `health < minThreshold`	3 lines of Kotlin
Database Evolution	`AfkDb.kt`	Room Migration 7 → 8 adding columns with default values	100% backward compatible

5. Phased Implementation Roadmap

Phase 1: Quick Wins (Immediate)

- Natural Camera Swivel & Sneak Pulse
- Minecraft Hearts & Hunger Drumsticks HUD
- Quick Command Macro Chips Bar
- Emergency Health Disconnect

Phase 2: Card & Connection Polish

- Live Ping & Player Count Pills on Cards
- 1-Tap Power Toggle on Cards
- Notification Action Buttons ([Stop All], [Reconnect])
- Intelligent 60s Server Reboot Delay
